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STELLAR WOBBLES

But there seems to be no compelling reason why such hypothetical... planets should not, in some instances, be much closer to their parent stars than is the case in the solar system. It would be of interest to test whether there are such objects.

— *Otto Struve, 1952*

2.1 At the Telescope

“Carolyn, we’re ready to move to the next target.” I’m talking to the Keck Observatory observing assistant, who is at the controls of one of the world’s largest telescopes, one of the twin 10-meter Keck telescopes. To be honest, it’s not even really necessary for me to make this request to Carolyn since she knows the routine of the California Planet Survey as well as I do, if not better. If the weather is good, we will methodically march down our target list, observing approximately one star every five minutes, for a grand total of about 120 stars in a single night. Before I can even finish my sentence, the control screen in front of me indicates that the telescope is on its way to the celestial coordinates of the next target of the night.

Many older astronomers blanch at the way that modern observational astronomy is conducted. For example, we are

sitting in an environmentally controlled room at sea level in Pasadena, California. We're working at a computer with three monitors that display every vital piece of information about the Keck 1 Telescope and the instrument we're using, the High-Resolution Echelle Spectrometer (HIRES). We're connected to the telescope across the Pacific Ocean via a high-bandwidth fiber-optic cable that mediates my conversation with the observing assistant, Carolyn. Unlike me, she is actually at the telescope, atop an extinct volcano called Mauna Kea on the Big Island of Hawai'i.

This is a far cry from the days when astronomers would travel to the mountaintop, put on heated bomber jackets and climb up to the prime focus of a telescope such as the Palomar 200-inch near San Diego. There at the top of the telescope, 40 feet above the dome floor in a tiny space-capsule-like box, the intrepid astronomer would spend the night observing. Perched in the observing "cage" for 9 to 12 hours with no breaks, the astronomer's job was to keep the night's target stars centered on the crosshairs of an eyepiece while sending fine guiding controls to the telescope via a wired, push-button controller. These were long, painstaking nights spent alone in the cold and darkness, without the ability to take breaks to walk around or stretch. As George Herbig once told me, "You'd take two thermoses with you up to prime focus. One empty and one full. At the end of the night, you'd bring two thermoses down: one empty and one full." There were no restroom breaks up at prime focus.

Nowadays we have computers to guide the telescope, and fiber-optic cables to carry a multitude of information about the telescope to observers sitting in a warm room at

sea level. Modern observational astronomers spend far less time and effort at the telescope than our earlier colleagues did. However, they spend just as much time planning their observing runs, analyzing their data and publishing their results. Computers rather than humans move the telescope across the sky, and the images and spectra are recorded on digital detectors rather than photographic plates. But the data still require a human interpretation, and the knowledge we glean from those data are just as critical as ever for our understanding of the Universe.

On this particular night, I am excited about star number 96 on our target list, also known as HD 94834, the 94,834th star in the Henry Draper Catalog of bright stars. Admittedly, it's a fairly nondescript star in the constellation Leo. It is too faint to warrant a Bayer or Flamsteed designation, like α Cen A or 70 Virginis. A search for the star in the astronomical literature shows that it appears in a catalog of stellar distances, and a catalog of stellar magnetic activity. However, to me this star is special because it affords an opportunity to see planets that were hidden from view during most of its prior lifetime. I started observing it back in the fall of 2007 as part of my search for planets around stars more massive than the Sun. HD 94834 reached the end of its main-sequence life about 10 million years ago—just yesterday in astronomical terms. On the main sequence, it “worked a nine-to-five job” generating the energy needed to hold itself up by fusing hydrogen in its core.

After about 4.6 billion years HD 94834 entered into “retirement” following its life of fusing hydrogen in its core on the main sequence. Having exhausted its source of

hydrogen fuel, the core began to contract. Paradoxically, as the core contracted, the rest of the star expanded in order to maintain hydrostatic equilibrium, and as a result, the star cooled. More important to me, the star also dramatically slowed its rotation,¹ thereby bringing the absorption lines in its spectrum into better focus. As part of my planet search program, I monitor the velocities of my target stars looking for accelerations along our line of sight that belie the existence of previously unknown planets. As we will see later in this chapter, these accelerations are observed as tiny shifts in the absorption spectra of the star due to the gravitational tug of an unseen planet. While on the main sequence, HD 94834 was rotating two orders of magnitude more rapidly than it does today. This rapid rotation smeared out its absorption lines, thereby obscuring information about the star's planets. But in retirement, the star has become an optimal target for my planet-search program.

The first three velocity measurements I made of HD 94834 back in 2007 and 2008 were completely flat, and initially the star appeared to be moving at a constant velocity with respect to the Solar System. However, the fourth velocity measurement made in the winter of 2008 showed a deceleration as the star's velocity decreased by 35 m s^{-1} , at a speed where it appeared to remain for the next year.

¹The star slowed in part because of the conservation of angular momentum as it expanded its radius. Another reason is that its magnetic field interacted with the stellar wind, allowing material to be carried along the magnetic field lines to large distances from the star. This transports angular momentum away from the star, allowing it to dramatically slow its rotation.

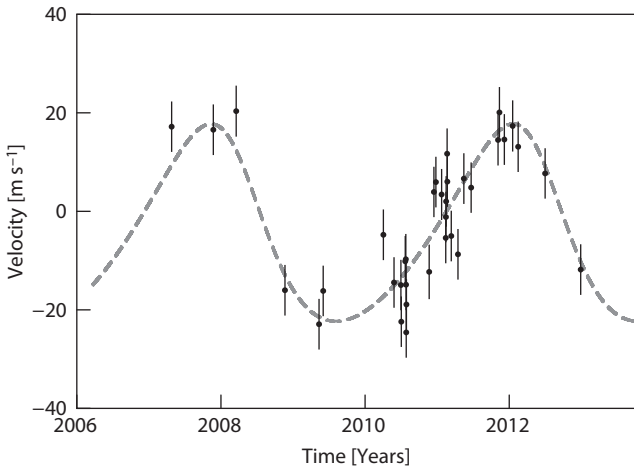


Figure 2.1. Radial velocity measurements of HD 94834 made at Keck Observatory with the HIRES spectrometer. The error bars represent the uncertainty due to instrumental errors and astrophysical “jitter” (see Section 2.5). The dashed line shows the best-fitting orbit model. Later in this chapter I demonstrate that the properties of this planet (minimum mass, period, eccentricity and semimajor axis) can be read by eye.

Then in 2010 the star began to noticeably accelerate once again, steadily changing its velocity by about 17 meters per second per year ($\text{m s}^{-1} \text{ yr}^{-1}$) until it returned to its 2008 velocity. In 2012 I noticed that it began to decelerate once again, making it a candidate for exoplanet discovery. However, at the moment it is only a possible planet that awaits confirmation from additional observations. Tonight at the telescope I am hoping that the photons that hit our detector carry information about the candidate planet’s orbit. If so, I will have discovered a

new planet in a 4.3-year, slightly eccentric orbit around the subgiant star HD 94834. The planet will be named according to tradition: HD 94834 b. The backstory of how astronomers identify and measure the properties of distant planetary systems is tied to the physics of orbital motion, which is the subject of the following section.

2.2 For Every Action

From our view within our Solar System one could ignore the fact that planets orbit the Sun and one would still be able to detect them and study their physical structure. Indeed, for thousands of years humans studied the night sky's wandering bodies with little to no knowledge of the gravitational bond between the Solar System planets and the Sun. Planets were the bright, nontwinkling objects that moved differently throughout the year, in contrast to the predictable motion of the stars.

However, to find planets orbiting other suns, it is crucial to have a correct understanding of the proper relationship between stars and their planets. Planets like those in our Solar System are gravitationally bound to stars, and as they orbit they can affect the light emitted from stars in different ways, depending on the geometric configuration of the system with respect to the observer. Perhaps the most general effect is the planet's gravitational tug on its star.

Most people who have taken a basic physics course know that stars exert a gravitational tug on their planets causing them to stay in orbit. However, it is not strictly correct to picture the planet orbiting a stationary star. This

is because, as Newton posited, every action has an equal and opposite reaction. As the star exerts a gravitational tug on its planet, keeping it in orbit, the planet tugs back on the star with a force that is equal and opposite of the star's gravitational pull. However, since the planet is so much less massive than the star, and because the force is equal to the mass of an object times its acceleration ($\vec{F}_{\text{net}} = m\vec{a}$), the resulting acceleration of the star is very small because its mass is very large.

In the following sections I derive the magnitude of a star's motion under the simplifying assumption of circular orbits, discuss the scaling relationships between the properties of the star's observed motion and the physical properties of the planetary system, and offer a brief overview of the methods used to measure the velocities of stars to high precision.

2.2.1 Radial Velocity Variations for a Circular Orbit

The famous seventeenth-century astronomer Johannes Kepler found that the planets in the Solar System move on elliptical (eccentric) orbits. However, it is instructive to first consider the special case of circular orbits (zero eccentricity, $e = 0$), since this case embodies most of the basic physics of orbital motion. Later we revisit the equations derived for circular motion and include the modifying effect of nonzero eccentricity.

Consider a snapshot of a star and its planet in a circular orbit, as viewed from above the system, i.e., pole-on (Figure 2.2). The star and planet will lie along a straight

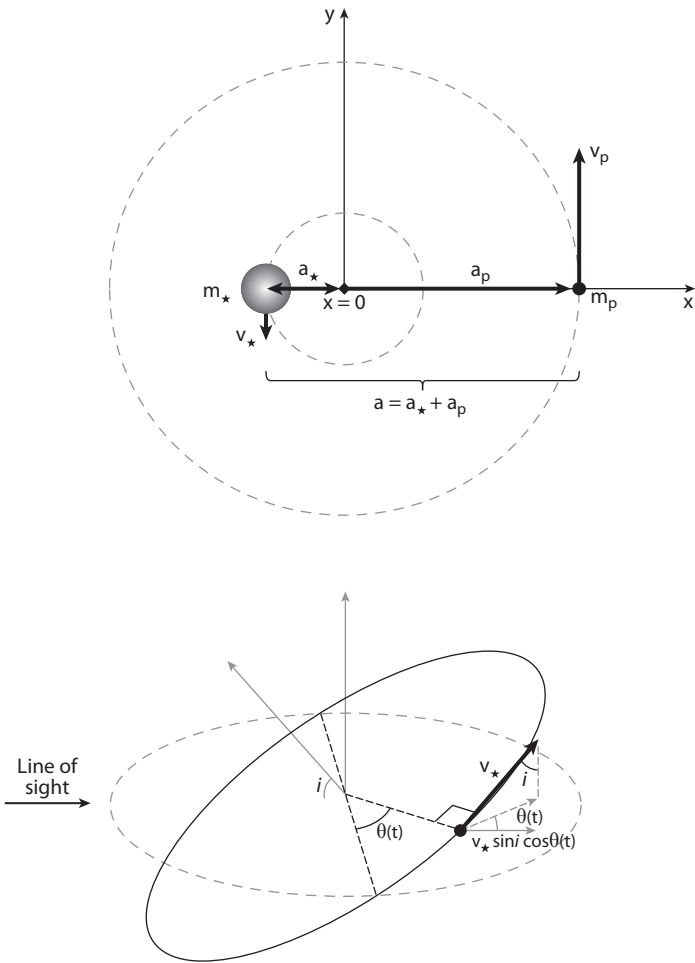


Figure 2.2. *Top:* Pole-on view of the star-planet orbital system. The star and planet orbit their mutual center of mass, or barycenter, which corresponds to $x = 0$, with the x -axis running from left to right. The planet's distance from the center of mass is given by $x = a_p$, while that of the star is a_* . The mean semimajor axis is

line, with velocities perpendicular to this line. The point at which the opposing gravitational forces are balanced between the star and planet is the center of mass, or barycenter of the system. Rather than planets orbiting their star, the actual situation involves the star and its planets orbiting their mutual barycenter, which we'll assume to be at rest with respect to the observer.²

The center of mass of two objects is placed at the origin of the x -axis, as in Figure 2.2, and the objects have masses $M_\star$ and m_p , for the star and planet, respectively. If $a_\star$ and a_p denote the distances of the star and planet, respectively,

²The barycenter of exoplanetary systems are generally moving with respect to the Solar System and other stars, because stars in the Galaxy have random motions with respect to one another.

Figure 2.2. (*Continued*)

defined to be $a = a_\star + a_p$. The semimajor axes are related by $a_\star = (m_p/M_\star)a_p$ and the velocities are related by the mass ratio, such that $v_\star = (m_p/M_\star)v_p$. *Bottom:* An illustration of the star's orbit. Unlike in the top panel, we are viewing the star's circular orbit at an arbitrary orientation in space, with a nonzero inclination i along our line of sight. Pole-on orbits correspond to $i = 0$ degrees and edge-on orbits have $i = 90$ degrees. The radial component of the star's velocity is given by the dot product of the velocity vector $\vec{v}_\star$ and the unit vector pointing from the observer (at left) to the orbit's barycenter, with an observed radial velocity $v_{\text{rad}} = v_\star \sin i \cos \theta(t)$. Positive velocities are away from the observer, and negative velocities correspond to motion toward the observer. The maximum velocity is reached at $\theta(t = 0) = 1$, corresponding to $K = v_\star \sin i$, which is the amplitude, K , of the RV signal caused by a planet with a minimum mass $m_p \sin i$. If $i = 0$ degrees, then $\sin i = 0$ and there is no component of the star's motion along the line of sight.

from the barycenter, we can make a choice of the origin and set the location of the barycenter to $\bar{x} \equiv 0$. The equation for the center of mass is given by

$$(2.1) \quad \bar{x} = \frac{-M_{\star}a_{\star} + m_p a_p}{M_{\star} + m_p} = 0$$

where the sign on the first term in the numerator is negative because the star is at a negative distance from the origin, $x_{\star} = -a_{\star} < \bar{x}$, whereas the planet is at a positive distance $a_p > \bar{x}$. Note that in this context, $a_{\star}$ and a_p are positions, with units of distance, and should not be confused with acceleration, which is denoted by the same lowercase a symbol. Additionally, we define the “mean semimajor axis,” $a = a_{\star} + a_p$. This quantity will reappear further along when we derive Newton’s version of Kepler’s Third Law.

Solving Equation 2.1 for the star’s semimajor axis gives

$$(2.2) \quad a_{\star} = \frac{m_p}{M_{\star}} a_p$$

which shows that the star’s distance from the center of mass is proportional to the planet’s distance times the planet-star mass ratio. Since the planet’s mass is so much less than the star’s mass ($m_p \ll M_{\star}$), $a_{\star}$ will be much smaller than a_p .

The relationship between a_p and $a_{\star}$ provides a relationship between the speeds of the planet and star. Note that the star and planet traverse the circumference of their orbits, $2\pi a_{\star}$ and $2\pi a_p$, respectively, once every orbital period P . Since speed is the distance traveled divided by

the time needed for the trip, this leads to

$$(2.3) \quad v_{\star} = \frac{2\pi a_{\star}}{P} = \frac{2\pi \left(\frac{m_p}{M_{\star}} a_p \right)}{P} = \frac{m_p}{M_{\star}} v_p$$

This shows that the speeds of the star and planet are also related by the mass ratio $m_p/M_{\star}$. Because the ratio $m_p/M_{\star}$ is very small, the speed of the star will be much less than the speed of the planet. This should agree with your physical intuition: the same force acting on a massive body (the star) will accelerate it less than if the force acted on a less massive body (the planet). Also note that $m_p v_p = M_{\star} v_{\star}$, which shows that the magnitude of the star and planet momenta are equal. This is to be expected when viewing the orbital system from the perspective of the center of mass: momentum is conserved.

Having expressions for the speeds of the star and planet is useful because a star–planet gravitational system can be considered a “virialized system.” As long as the system doesn’t change its shape (moment of inertia) over time baselines that are long compared to the orbital period, it can be adequately described by the average kinetic and potential energies of its constituents. Kinetic energy involves an object’s speed and mass of the gravitationally bound objects, and the potential energy is related to masses and separation of the two objects. An orbit satisfies the conditions of a virialized system, since the time-averaged shape of a two-body orbit is constant (the same holds for stable multibody orbits). Working with the energy of the system allows us to find a relationship among the period, semimajor axis and stellar mass. Of the many ways to

derive this relationship, I find the following derivation highly intuitive and simple, and familiarity with the virial theorem is useful for solving a wide range of astrophysical problems.

The virial theorem states that at all times the kinetic energy, $\mathcal{K}$, of a virialized system is half the potential energy holding the system together, U , or

$$\mathcal{K} = -\frac{1}{2}U$$

$$(2.4) \quad \frac{1}{2}m_p v_p^2 = -\frac{1}{2} \left(-\frac{Gm_p M_\star}{a} \right)$$

where I have replaced a_p with a since $m_p \ll M_\star$ and $a_p \gg a_\star$. Solving for v_p gives

$$(2.5) \quad v_p = \sqrt{\frac{GM_\star}{a}}$$

Note that this same expression can be derived by setting the centripetal acceleration v_p^2/a , equal to the gravitational force per unit mass, $GM_\star/a^2$.

Substituting $v_p = (M_\star/m_p)v_\star$, and $v_\star = 2\pi a_\star/P$ in Equation 2.5 and doing a bit of algebraic manipulation results in

$$(2.6) \quad P^2 = \frac{4\pi^2 a^3}{GM_\star}$$

This relationship is equivalent to Newton's version of Kepler's Third Law of planetary motion under the assumption that the star's mass is much larger than the planet's mass ($M_\star + m_p \approx M_\star$). If the period, P , is measured in

years, the semimajor axis a is in astronomical units (AU; where 1 AU is the Earth–Sun distance) and the star’s mass is equal to that of the Sun ($M_\star = 1 M_\odot$), then we can recover Kepler’s original Third Law,

$$(2.7) \quad P^2 \propto a^3$$

We now have all of the puzzle pieces necessary to understand the motion of the star in response to an orbiting planet. We have figured out how the speeds of the star and planet are related (Equation 2.3), and we have figured out the relationship among the period, semimajor axis and mass of the system, with the mass dominated by the central star $M_\star \gg m_p$ and $a_p = a$ (Equation 2.6).

Now let’s put it all together starting with the expression of the planet’s orbital speed in Equation 2.5. The planet’s speed is related to the star’s speed via the mass ratio, giving

$$(2.8) \quad v_\star \left(\frac{M_\star}{m_p} \right) = \left(\frac{GM_\star}{a} \right)^{1/2}$$

Since the semimajor axis, a , is not an observable quantity with the radial velocity technique, we can use Kepler’s Third Law to replace a with the planet’s orbital period, P , and the stellar mass, $M_\star$. After a bit of algebra, the star’s speed is given by the simple product of four quantities, one comprising various constants and the other three related to the physical properties of the system:

$$(2.9) \quad v_\star = (2\pi G)^{1/3} M_\star^{-2/3} P^{-1/3} m_p$$

If we express the stellar mass, period and planet mass in terms of the mass of the Sun ($M_\odot$), the period in years,

and the planet mass in Jupiter masses (M_{Jup}), respectively, we can recover the useful, quantitative relationship

$$(2.10) \quad v_{\star} = [28.4 \text{ m s}^{-1}] \left(\frac{M_{\star}}{M_{\odot}} \right)^{-2/3} \times \left(\frac{P}{\text{year}} \right)^{-1/3} \left(\frac{m_p}{M_{\text{Jup}}} \right)$$

Thus, a Jupiter-mass planet in a one-year orbit around a solar-mass star will cause its star to move at a speed of 28.4 m s^{-1} . To put this in perspective, 1 meter per second is about 2.2 miles per hour. A Jupiter-sized planet in a one-year orbit would cause the star to move about 62.5 miles per hour. This is decently fast for a car, but it is extremely slow by astrophysical standards. For example, the Sun orbits the center of the Milky Way Galaxy at a speed of 486,000 miles per hour (roughly 220 km s^{-1}), which is typical for stars in the Galaxy. The velocity variations induced by planets are a tiny factor on the movement of a typical star.

We can also express the star's speed in terms of the more physical variable a , the semimajor axis, rather than the period P . This can be done using Kepler's Third Law to replace P , resulting in

$$(2.11) \quad v_{\star} = [28.4 \text{ m s}^{-1}] \left(\frac{M_{\star}}{M_{\odot}} \right)^{-1/2} \times \left(\frac{a}{1 \text{ AU}} \right)^{-1/2} \left(\frac{m_p}{M_{\text{Jup}}} \right)$$

If we instead express m_p in Earth-masses, the numerical factor in front becomes much smaller. Jupiter is about 318 times more massive than the Earth, so v_* becomes 318 times smaller, or 9 cm s^{-1} . Stop and think for a moment about this speed: 9 *centimeters* per second. The width of a piece of printer paper is about 20 centimeters. Trace your finger across the sheet such that it takes two seconds to go from one side to the other. This is how fast the Sun—a giant ball of gas, 100 times the diameter of the Earth—moves in response to the Earth’s gravitational tug. And somehow astronomers must detect the motion of this ball of gas moving at 10 centimeters per second from a distance of tens or hundreds of light-years in order to find Earth-mass exoplanets at 1 AU around their stars.

Throughout this derivation we have assumed that we are viewing the planet’s orbit edge-on, so the line-of-sight component of the stellar velocity, its radial velocity v_{rad} , is at its largest value. However, imagine a situation in which the orbit plane is parallel to the sky such that we are viewing the planetary system from “above,” or pole-on, rather than edge-on. In this case, the planet and star will always move perpendicular to our line of sight, and we would detect zero motion in the radial direction. This is because we see only the projection of the star’s velocity vector along our line of sight, and a face-on orbit has no portion of the star’s motion aimed toward or away from us.

If the star orbited by a Jupiter-mass planet with a period of 1 year were observed from afar with an edge-on orientation, the maximum velocity along the line of sight, given by $K \equiv \max(v_*)$, would be 28.4 m s^{-1} . The

observable, K , is often referred to as the radial velocity “semi-amplitude” of the orbit. As can be seen in Equation 2.10, K is directly proportional to the planet’s mass: $K \propto m_p$. However, since we do not typically know the inclination of the orbit, the mass of the planet is degenerate with $\sin i$, and the value of m_p inferred from the amplitude represents the planet’s *minimum mass*, expressed as $m_p \sin i$. If the inclination is smaller, then $\sin i$ is smaller, and the planet’s true mass is larger than one would measure from K .

Next, we must account for the time variability of the star’s radial velocity. The previous expressions give the star’s *speed*, or equivalently the amplitude of the time-variable motion along the line of sight. But for a circular orbit the line-of-sight component of the star’s velocity will vary along the orbit as a cosine wave of the form

$$(2.12) \quad v_{\text{rad}}(t) = K \cos [\theta(t) - \omega]$$

where

$$(2.13) \quad \theta(t) = \frac{2\pi(t - T_p)}{P}$$

I have added two important terms that relate to the phase of the planet’s orbit. The first is ω , the *argument of periastron*, which describes the rotation of the orbit with respect to the observer. For an eccentric orbit, periastron corresponds to the planet’s closest approach to the star, while apastron is the maximum distance between the star and planet.³ For a circular orbit there is no actual periastron, so ω can be arbitrarily set to zero, which is

³As a mnemonic device, I think of “peri” as being my friend, who I want close. Apastron starts with “a,” like “away.”

defined as the point along the orbit when the angle $\omega(t)$ is such that the line connecting the star and the barycenter is perpendicular to the line of sight, with the star moving away from the observer, as shown in Figure 2.2. As we'll see later, ω can be nonzero for eccentric orbits, since an elongated orbit can be oriented with an arbitrary angle with respect to the observer's line of sight.

The second newly introduced term is the time of periastron passage, T_p . This term sets the phase of the orbit. Even though we have defined $\omega = 0$ for a circular orbit, the observed sinusoidal motion of the star's radial velocity will be random—different orbital systems will reach their maximum radial velocities at different times. For this reason T_p corresponds to the time when the planet passes through periastron.

The time between consecutive peaks gives the orbital period P . When combined with the amplitude of the signal, K , Equation 2.10 can be used to measure the minimum mass of the planet, $m_p \sin i$, divided by the mass of the star raised to the $2/3$ power, or $m_p M_\star^{-2/3}$. Thus, in order to estimate the planet's minimum mass, one needs to know the star's mass. This is a theme that is repeated throughout this book: to understand the physical properties of planets, one must understand the physical properties of the stars they orbit.

2.3 Eccentric Orbits

To keep things fairly intuitive and easy to follow, up to this point I have neglected the effects of eccentricity throughout the derivation of the radial velocity of a star

with a planet. Doing this is fine for getting a handle on the basic scaling of the problem, but as Kepler's First Law states, planets move in elliptical orbits. And while most of the planets of the Solar System have nearly circular orbits with $e < 0.1$ —with the exception of Mercury, which has $e = 0.2056$ —exoplanets have been discovered with a wide range of eccentricities varying from nearly circular up to cometlike orbits with $e = 0.934$ (HD 80606 b; Naef et al. 2001).

Including orbital eccentricity, e , also modifies the equation for the Doppler amplitude, K , through a multiplicative term involving e :

$$(2.14) \quad K = 28.4 \text{ m s}^{-1} \left(\frac{M_{\star}}{M_{\odot}} \right)^{-2/3} \left(\frac{P}{\text{year}} \right)^{-1/3} \times \left(\frac{m_p \sin i}{M_{\text{Jup}}} \right) (1 - e^2)^{-1/2}$$

The time variability of the radial velocity of a star, v_{rad} , is given by

$$(2.15) \quad v_{\text{rad}}(t) = K(\cos[\theta(t) - \omega] + e \cos \omega)$$

However, in an eccentric orbit $\theta(t)$ does not vary linearly with time. Instead, $\theta(t)$ is related to a new variable called the *eccentric anomaly*, $E(t)$, by

$$(2.16) \quad \tan \frac{\theta(t)}{2} = \left(\frac{1+e}{1-e} \right)^{1/2} \tan \frac{E(t)}{2}$$

When $e = 0$, Equation 2.16 reduces to $\cos E(t) = \cos \theta(t)$, which is equivalent to $E(t) = \theta(t)$, where $\theta(t)$ is given by Equation 2.13 and includes the time of periastron passage, T_p , and the argument of periastron, ω .

The eccentric anomaly can be computed at a time t and an eccentricity e through the transcendental equation

$$(2.17) \quad E(t) = e \sin E(t) + \frac{2\pi(t - T_p)}{P}$$

At this point, why we did the derivations for the much simpler case of a circular orbit should be clear. Whereas a proper derivation requires consideration of the eccentricity, its derivation involves more geometry and algebra than physical insight. In practice, solving for the velocity of a star as a function of time can be accomplished with the following algorithm:

- Start with an initial time, t_0 , in Equation 2.17 and numerically solve for $E(t_0)$ using an iterative procedure, such as a Newton-Raphson scheme.
- Solve for $\theta(t_0)$ by plugging e and $E(t_0)$ into Equation 2.16.
- Use $\theta(t_0)$ to compute $v_{\text{rad}}(t_0)$ using Equation 2.15.
- Repeat for a new time, $t_1 = t_0 + \Delta t$, where Δt is a time increment that is small compared to the orbital period.

The need to solve a transcendental equation makes computing Keplerian orbits fairly slow. Every ten years or so someone devises an improved method of solving Kepler's equations, which allows for more efficient

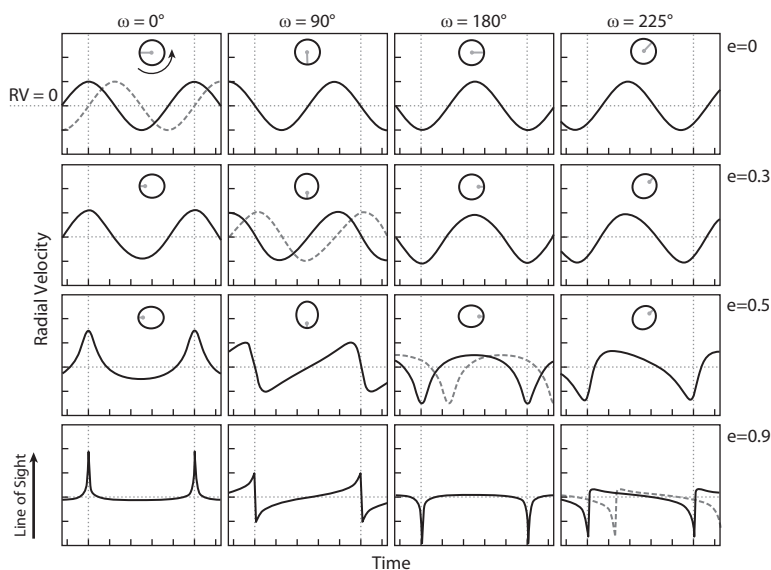


Figure 2.3. Illustration of the effects on the radial velocity of a star with varying the eccentricity e and argument of pericenter ω . The small orbit at the top of each panel shows the orbital motion of the *planet* with a period P (arbitrary units and magnitude), eccentricity e , orientation ω and phase T_p . The gray line in each orbit configuration indicates the angle ω with respect to the line of sight to the system (the angle ω is also labeled for each column along the top of the figure). The solid line showing the RV variation

simulations of N-body gravitational problems (Wisdom & Holman, 1991), and faster fitting of radial velocity time series through partially linearizing the equations (e.g., Wright & Howard, 2009) or through novel use of computing hardware (Ford, 2009).

Example orbits with different eccentricities, e , and phases (ω and T_p) are illustrated in Figure 2.3. In each panel, the planet's orbit, shown pole-on, is illustrated, with

the observer assumed to be viewing the system from a vantage point at the bottom of the page. From this vantage point, the hypothetical observer will see the orbit edge-on. The vertical dotted line on the left in each panel denotes $t = 0$. Different values of ω rotate the orbit with respect to the observer.

Note that the corresponding radial velocity plots in each panel correspond to the star, not the planet. The planet and the star move counterclockwise in each panel. The diagonal panels, moving from upper left to lower right, illustrate the effect of changing T_p by a quarter of the orbital period. This has the effect of shifting the radial velocities to the right.

A great deal of information is contained in this figure, and understanding all aspects of it will not be achieved immediately. However, taking time to think through each example will help the student gain a physical intuition for reading radial velocity time series.

Figure 2.3. (*Continued*)

in each panel corresponds to the radial motion of the *star* (not the planet, which is shown above). Note that the stellar orbit that gives rise to the RV signal has a sign opposite to that of the planetary motion, corresponding to a phase difference of π . The times $t = 0$ and $t = P$ are indicated by the vertical, dotted lines. The gray, dashed curve is the same RV signal with the time of periastron changed to $T_p = P/4$, demonstrating how T_p governs the phase of the periodic RV variations. The rows show orbits with constant e as indicated on the right of the figure, and the columns are for constant ω as indicated along the top of the figure. In the first three rows, the orientation and shape of the orbit of the planet around the central star (gray circle, not to scale) are shown, with the orbit viewed from the bottom of the page.

2.3.1 Example: Reading Radial Velocities

Figure 2.1 shows the radial velocity (RV) measurements of HD 94834 made with the HIRES instrument at the Keck Observatory, along with the best-fitting orbit model shown as a dashed line. We can estimate the period by measuring the distance between consecutive peaks, which gives $P \approx 4.3$ year. The Doppler amplitude K is given by the distance from $RV = 0$ to the maximum value, or $K \approx 20 \text{ m s}^{-1}$. Making the assumption of a circular orbit, and using the approximate stellar mass $M_\star = 1.3 M_\odot$, we can estimate the minimum planet mass by solving Equation 2.10 for $m_p \sin i$:

$$\begin{aligned} \frac{m_p \sin i}{M_{\text{Jup}}} &= \left(\frac{K}{28.4 \text{ m s}^{-1}} \right) \left(\frac{M_\star}{M_\odot} \right)^{2/3} \left(\frac{P}{\text{year}} \right)^{1/3} \\ (2.18) \quad m_p \sin i &= \left(\frac{20}{28.4} \right) (1.3)^{2/3} (4.3)^{1/3} \\ &\approx 1.4 M_{\text{Jup}} \end{aligned}$$

Using Kepler's Third Law of motion (Equation 2.6) we can also solve for the planet's semimajor axis:

$$\begin{aligned} a &= \left(\frac{GM_\star}{4\pi^2} \right)^{1/3} P^{2/3} \\ (2.19) \quad &= [1 \text{ AU}] \left(\frac{M_\star}{M_\odot} \right)^{1/3} \left(\frac{P}{1 \text{ year}} \right)^{2/3} \\ &= (1.3)^{1/3} (4.3)^{2/3} \text{ AU} \approx 2.9 \text{ AU} \end{aligned}$$

2.4 Measuring Precise Radial Velocities

Astronomers measure the velocity of stars using the Doppler shift induced in a star's light as it moves. We encounter the Doppler effect in our daily lives as a car approaches, passes, and then drives away from us. At first the engine sound is high pitched, and it gradually sounds lower pitched as it passes and then drives away from us. This is because the sound waves are compressed as the car approaches, and then stretched as it recedes. Since light can be described as a wave, the Doppler effect will cause light waves to become compressed ("blueshifted") as a star moves toward the Earth, and stretched ("redshifted") as the star moves away.

Despite how it is usually depicted in widely used illustrations, the Doppler technique of finding planets does not involve first looking for blue light, and then looking for red light as the star moves in different directions. At all times the distribution of energy emitted from a star is to first-order a blackbody, or Planck distribution, dependent only on the star's surface temperature. However, inspection of a star's spectrum shows deviations from that of a perfect, smooth blackbody emitter. These deviations are due to the absorption and emission of photons by atoms and molecules in the star's atmosphere, and the positions of these "lines" encode information about the star's motion.

The centers of stellar absorption lines correspond to the quantum levels of atoms and molecules in the star's atmosphere, and photons on their way out of the star encounter various atoms and molecules. If the wavelength of a photon corresponds to the energy difference between

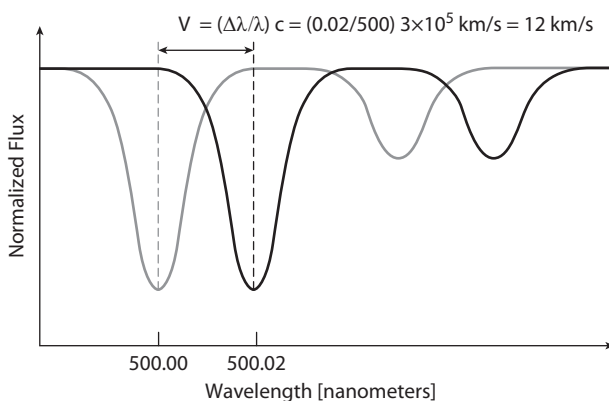


Figure 2.4. Illustration of a stellar spectrum at rest (gray) and the same spectrum Doppler-shifted (black). The change in wavelength between the two spectra is $\Delta\lambda = 0.02$ nm, and the center of the line was originally located at $\lambda = 500$ nm. The ratio of the change in the line's position to its original position, when multiplied by the speed of light ($c \approx 3 \times 10^5$ km s⁻¹), yields the star's velocity. In this example, the Doppler shift is caused by the star changing its velocity by 12 km s⁻¹.

quantum states of an atom or molecule, the photon can be absorbed. Thus, there are well-defined central wavelengths for every absorption line.⁴ If the star moves with respect to us, we'll see the wavelengths of the absorption lines change due to the Doppler effect.

A star's absorption lines can be observed using a spectrometer, which measures the amount of light (flux) from a star in small wavelength intervals. For a typical

⁴Absorption lines have widths because of various line-broadening mechanisms. The primary broadening sources in most stars are due to stellar rotation and the motion of hot and cold cells of gas in stellar atmospheres.

spectrometer used to detect planets, light is measured in intervals of about 0.6 nanometer (nm). This might seem small, but consider the wavelength shift of a typical absorption line caused by a Jupiter-sized planet in a one-year orbit. The Doppler shift, z , is given by

$$(2.20) \quad z = \frac{\Delta\lambda}{\lambda_0} = \frac{v_\star}{c}$$

where $c = 3 \times 10^8 \text{ m s}^{-1}$ is the speed of light, λ_0 is the wavelength where the stellar absorption line is normally centered, and it is shifted by an amount $\Delta\lambda$. Here, I have assumed that the star's motion is negligible compared to the speed of light ($v_\star \ll c$).

For a star moving at $v_\star = 30 \text{ m s}^{-1}$, a line normally at $\lambda = 600 \text{ nm}$ will be shifted by $\Delta\lambda = 6 \times 10^{-5} \text{ nm}$. This is about 10^5 or 10,000 times smaller than the wavelength spanned by a single detector element of a typical spectrometer. Fortunately, every one of the thousands of lines in a star's spectrum encodes the exact same Doppler shift. By measuring the average shift of all of these lines, astronomers can attain the precision necessary to detect planets as small as several Earth-masses in periods of a few days using existing instruments. Future instruments will have even better sensitivity.

However, attaining this level of precision is not trivial. One must not only measure a shift $\Delta\lambda$, but the shift must be measured with respect to a reference wavelength λ_0 . Thus, the measurement of highly precise radial velocities relies on a precise mapping of wavelength to pixel within the spectrometer. The spectrometer is a complicated

optical instrument. However, it is essentially a prism to disperse light, and at the end of the light path is a simple charge-coupled device (CCD), on which the star's spectrum is recorded for later analysis. The CCD is similar to the detector in a commercial digital camera, and it comprises a grid of small (~ 15 micron) pixels. Different wavelengths of light are directed onto different pixels. A closely spaced pair of absorption lines can be distinguished as individual lines only if they are separated by more than this resolution element.

The primary problem is the spectrometer is a moving platform. Changes in the ambient temperature surrounding the spectrometer can change the position of optics by a fraction of a wavelength. Changes in pressure can change the index of refraction of the air near the optics. And vibrations can shift the detector with respect to the light path. All of these effects change the mapping of wavelength to pixel within the spectrometer. These changes in the wavelength mapping can easily be mistaken for a Doppler shift of several kilometers per second, which is plenty to mask the much smaller shift due to a planet.

One way to mitigate these false Doppler shifts is to impose a locally generated absorption spectrum onto the stellar spectrum. This can be accomplished by placing a clear, Pyrex glass cell containing a molecular gas at the entrance of the spectrometer. Since the gas cell is not moving, the narrow molecular absorption features serve as a stationary grid, mapping wavelength to pixel, against which the motion of stellar absorption lines can be measured to high precision. While the spectrometer optical components might shift throughout the night, the

locations of the lines produced by the gas cell remain fixed in wavelength. This is the technique used at the Keck Observatory with the HIRES spectrometer.

The other method of measuring highly precise radial velocities is to ensure that the instrument does not move. The absorption-cell technique is most often used as a post hoc solution to intrinsic instabilities of general-use spectrometers. However, if a spectrometer is built specifically for precise radial velocities, it can be designed with stability in mind. This stable-platform approach is used on the High-Accuracy Radial-velocity Planet Search (HARPS) spectrometer. HARPS is housed below its telescope and it is enclosed in a temperature- and pressure-controlled environment. The wavelength scale is periodically checked with a lamp that emits light at known, discrete wavelengths. However, because of the intrinsic stability of HARPS, the wavelength scale remains static over the course of a night. For stars requiring the highest possible precision, a separate fiber-optic cable can shine calibration light at the same time that the stellar spectrum is recorded.

2.5 Stellar Jitter

Even with a perfect instrument, the detection of low-mass planets will be complicated by the nonstable nature of stars. While stars can be thought of as spherical black-bodies, in reality they are large balls of gas with no true surface. The “surface” is known as the photosphere, which marks the radius within the star where the gas transitions

from opaque to translucent to outgoing photons. This pseudo-surface, known as the photosphere, can be jostled about by the bubbling, convective gas just below it. The rising convective cells then cause the star to ring at natural vibrational modes that show up as periodic variations in the velocity of the stellar surface. However, since the surface of the star cannot be resolved, all of this jostling results in radial velocity noise that is astrophysical in nature, rather than originating in the instrument. This astrophysical noise is known as “stellar jitter.”

In addition to noise sources in the interior of the star, the surface of the star is not uniformly bright. Instead, the surface of stars like the Sun are dotted by dark star spots, bright regions called plage, and the occasional flare. Since stars are approximately spheres, they appear to an observer as a circular disk projected on the sky, with half of the surface approaching the observer and half of the surface receding, due to stellar rotation. The net effect of rotation on an unblemished stellar surface would be for the two halves of the star, one redshifted and one blueshifted, to balance out and no net apparent velocity would be produced. Instead, the star’s absorption lines appear to be broadened by the redshifted and blueshifted portions of the stellar surface.

Blemishes on the stellar surface will tip this balance and result in a nonzero net velocity across the stellar surface and absorption lines. For example, a spot on the blueshifted hemisphere of the star would result in a net redshift. Then, as the spot moves across the star’s surface following the star’s rotation, the net shift would change to a blueshift. This change in the star’s surface velocity can be mistaken

for the Doppler signal of a planet, with an orbital period equal to the star's rotation period.

The first method of dealing with jitter is to just treat it as an additional source of noise. Even though the processes that give rise to artificial radial velocities produce signals that are time-correlated, if the signal is sampled sporadically—as usually happens due to the vagaries of telescope scheduling—the jitter will look like random scatter. Various calibrations indicate what level of jitter one can expect for a star of given properties. This expected level of jitter can simply be added in quadrature to the measurement uncertainties of the RV measurements, resulting in inflated error bars for each data point.

A better method of dealing with jitter is to use a physical model to fit the jitter signal along with the planet signal. However, this method can only be used when the jitter “signal” is time-resolved with a dense set of measurements spanning the typical timescale for jitter to manifest itself and evolve. For example, jitter due to star spots modulates the RV signal of the star on a timescale of the star's rotation. For Sun-like stars, this rotation timescale is tens of days (24 days for the Sun). An additional feature of this spot-induced jitter is that it is ephemeral, as opposed to the constant signal from the planet. This is because spots come and go on monthly timescales. Thus, if one can sample the star's RV once per night, every night over months, one can in principle see the spot modulation oscillating and fading in and out, along with the underlying, permanent planet signal. For more on this method, I refer the reader to the work of Xavier Dumusque and collaborators (2012), who used high-cadence monitoring of α Cen B to detect a

planet (candidate) with a minimum mass ($M_p \sin i$) close to the Earth's mass and that has a Doppler signal below the jitter level of the star.

2.6 Design Considerations for a Doppler Survey

In the previous sections I provided the basic physical picture of a planet tugging on its star, thereby producing a Doppler shift in the star's spectrum. I also demonstrated that the expected signal is quite small, yet measurable with modern spectrometers. In this section I provide examples of how one would design a survey to detect two types of planets. The first case is close-in, gas-giant planets known as "hot Jupiters." These are the easiest planets to detect using the Doppler technique. However, hot Jupiters are found around only $\sim 1\%$ of stars. Thus, the survey approach must maximize the number of stars searched in order to have a reasonable expectation of finding planets.

The second case focuses on Earth-mass planets in the habitable zones of their stars. Compared to hot Jupiters, Earth-sized planets are common throughout the Galaxy. However, because the Doppler amplitude scales proportionally to the planet's mass and inversely with the semimajor axis ($K \propto a^{-1/2} m_p \sin i$), the signals from these planets will be orders of magnitude smaller than the Doppler variations induced by hot Jupiters. Thus, such a survey must be optimized in a very different manner than one searching for more massive planets.

2.6.1 Example 1: Searching for Hot Jupiters

According to Equation 2.10, a Jupiter-mass planet in a 1-year orbit around a Sun-like star induces a signal with an amplitude $K \approx 30 \text{ m s}^{-1}$. Hot Jupiters orbit much closer to their stars with periods of about 3 days, instead of 365 days. The amplitude scales as $P^{-1/3}$, so a hot Jupiter will induce a signal that is $(P_{\text{HJ}}/P_{\text{AU}})^{-1/3} = (3/365)^{-1/3} \approx 5$ times larger than a Jupiter at 1 AU, or $K \approx 150 \text{ m s}^{-1}$.

Since modern spectrometers can routinely measure stellar radial velocities with a precision of 3 m s^{-1} , a middle-aged ($>1 \text{ Gyr}$), Sun-like star with a hot Jupiter is readily distinguished from a star without one. Even with three to four RV measurements, a star with a hot Jupiter will exhibit RVs with a much larger scatter than it would if it were solitary. This means that only a few RV measurements are necessary to identify stars that are likely to harbor hot Jupiters.

This “quick-look” strategy was first implemented by Debra Fischer with her *Next 2000 Stars* (N2K) survey (Fischer et al., 2005). She and her team surveyed a large number of stars by scheduling telescope time in several three-night observing runs and observing their target stars once per night. For each star’s sequence of three RVs, the N2K team calculated the standard deviation of the points, σ_V (top panel of Figure 2.5). Stars that have σ_{RV} larger than some threshold were then followed up intensively to characterize the orbit of the planet candidate. This last step typically required 10–15 additional RV measurements. The N2K survey resulted in the discovery of 17 Jupiter-mass exoplanets, of which two were subsequently found

to transit (eclipse) their host stars, demonstrating the efficiency of the quick-look survey strategy. The lower panel of Figure 2.5 shows the RV time series for a planet I helped detect and announce when I was a member of the N2K team as a graduate student.

2.6.2 Example 2: Searching for Habitable-Zone Planets

The habitable zone (HZ) of a star is the range of semi-major axes at which a planet's equilibrium temperature is amenable to the presence of liquid water on its surface. The details of habitability are complicated and varied. Indeed, the topic deserves its own book, and I encourage the reader to see Jim Kasting's *How to Find a Habitable Planet* (2010).

For our purposes, we can think of the HZ as a semimajor axis a_{HZ} characterized by constant planetary equilibrium temperature, T_{eq} , around stars of various masses, $M_{\star}$. The equilibrium temperature of a planet is reached when the power it receives from the central star is equal to the power the planet emits as thermal radiation. The power per area absorbed at the location of the planet, also known as the flux F with units of energy per time per area, is

$$(2.21) \quad F_{\text{in}} = \frac{L_{\star}}{4\pi a^2}$$

where $L_{\star}$ is the luminosity of the star, or total power output. The planet absorbs energy on the day-side hemisphere, which has a projected area πR_p^2 as seen from

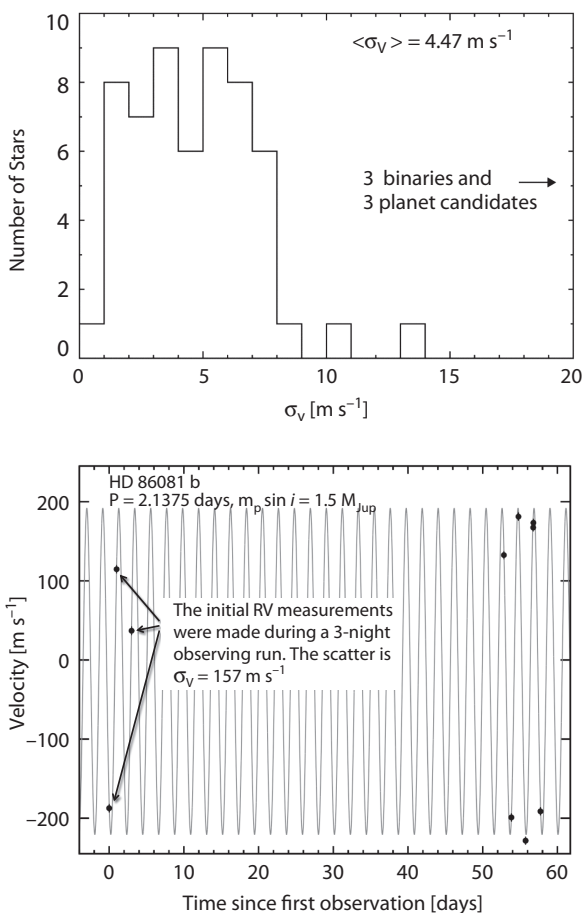


Figure 2.5. *Top:* The distribution of σ_V for the stars in the Keck N2K survey. σ_V is the standard deviation of the first 3–4 RV measurements of each target star. The mean level of RV scatter is only 4.47 m s^{-1} , indicating that most stars in the survey lack a Jupiter-mass planet. A few stars in the sample show very large σ_V , and these stars are subsequently followed up with 10–15 additional RV measurements to confirm and characterize the

the star. Thus, the energy absorbed per unit time by the planet is $\text{Area} \times (\text{Flux in}) = 4L_{\star}(R_p/a_{\text{HZ}})^2$. Here, I have ignored the amount of light reflected at the planet's surface.

The planet is approximately a blackbody of temperature T_{eq} , and thus also radiates flux over its entire surface, which has a spherical area $4\pi R_p^2$. The flux at the surface of a blackbody is $F_{\text{out}} = \sigma T_{\text{eq}}^4$, where σ is the Steffan-Boltzmann constant. The power output is $\text{Surface area} \times (\text{Flux out}) = 4\pi R_p^2 \sigma T_{\text{eq}}^4$. Equating the input and output power and solving for the a_{HZ} gives

$$(2.22) \quad a_{\text{HZ}} = \left(\frac{L_{\star}}{\pi \sigma T_{\text{eq}}^4} \right)^{1/2}$$

Since the HZ is characterized by a constant T_{eq} , the location a_{HZ} is dependent only on the stellar luminosity $L_{\star}$. For Sun-like stars a good approximation for the relationship between luminosity and mass is given by $L_{\star} \sim M_{\star}^4$, which leads to

$$(2.23) \quad a_{\text{HZ}}(M_{\star}) \sim M_{\star}^2$$

The Earth resides in the Sun's habitable zone at $a_{\text{HZ}}(M_{\star} = M_{\odot}) = 1 \text{ AU}$. A star with 70% the mass of the Sun has $a_{\text{HZ}} = 0.7^2 \approx 0.5 \text{ AU}$.

Figure 2.5. (Continued)

planetary orbit. *Bottom:* An example hot Jupiter discovered by the N2K Consortium. The first three points have $\sigma_V = 157 \text{ m s}^{-1}$, indicative of a planet candidate. The planet was later confirmed and published as HD 86081 b (Johnson et al., 2006).

We can now derive the velocity amplitude of a HZ planet K_{HZ} using Equation 2.11, which results in

$$(2.24) \quad K_{\text{HZ}}(M_{\star}) \sim M_{\star}^{-3/2} m_p \sin i$$

This result indicates that for a given planet mass in the habitable zone, the largest Doppler signals are expected for stars with lower masses since the amplitude scales inversely with $M_{\star}$. For example, an Earth-sized planet orbiting at a_{HZ} around a red dwarf with $M_{\star} = 0.3 M_{\odot}$ will have a Doppler amplitude that is six times larger than the same planet in the HZ of a Sun-like star, or $K_{\text{HZ}} \approx 55 \text{ cm s}^{-1}$, compared to 9 cm s^{-1} for the Sun orbited by the Earth. Of course, our calculation of the location of the habitable zone includes several implicit, simplifying assumptions. A fuller treatment would consider the effect of the star's spectral energy distribution on the planet's temperature (e.g., Kasting et al., 1993; Shields et al., 2013). However, this simple derivation demonstrates that stellar mass is clearly an important consideration in planning a search for HZ planets.

2.7 Concluding Remarks

In this chapter we examined how planets belie their presence by gravitationally tugging on their host stars, which causes stars to move back and forth along our line of sight. Astronomers can detect this motion by monitoring the velocities of stars using spectrometers to track the Doppler shift of stellar absorption lines. Because the amplitude

of the RV variability scales as $K \propto a^{-1/2}$, and because planets have periods less than the observing baseline, the Doppler technique is most sensitive to close-in planets. Presently, the longest-running Doppler surveys have time baselines of about 20 years, and the planet with the longest, precisely measured orbital period is 55 Cnc d, which has $P = 13.44 \pm 0.08$ years.

The majority of planets detected with the Doppler technique have periods less than a year. In fact, one of the shortest-period planets also orbits the star 55 Cnc. The planet, 55 Cnc e, has a period of only 17.76 *hours*! The planet's mass is $m_p = 8.3 M_\oplus$, which is unambiguously measured because the inclination is known very precisely: $i = 83.4^{+1.5}_{-1.7}$ degrees. The inclination is known because the planet passes in front of the central star as viewed from Earth. The details of these planet transits are the subject of the next chapter.